

OBSTRUCTIONS TO WEIL POSITIVITY: A CATALOGUE OF NO-GO THEOREMS, FALSIFIERS AND COUNTERMODELS IN THE PRIME-GAMMA DOOB COORDINATE

DAVID ALEJANDRO TREJO PIZZO

ABSTRACT. The companion paper reduces the Riemann Hypothesis, along the Connes–Consani–Moscovici and Suzuki route, to a single strict inequality on a cofinal family of finite rows: the physical source surplus $\mathfrak{g} > \delta$. This paper does not attempt that inequality. It does the complementary and, we argue, currently more useful thing: it records in one place the mechanisms which provably cannot prove it.

We work throughout in the full-kernel Doob coordinate, in which Weil positivity is the single sharp Poincaré inequality $\mathcal{E}_K(r) \geq \frac{1}{2} \text{Var}_{\mu_K}(r)$, with $\mathcal{E}_K$ the positive jump energy retaining every ordinary von Mangoldt atom and the complete Gamma jump measure. The catalogue contains thirty-one entries in eight families: order and total positivity; coupling, transport and martingale routings; local squares, sums of squares and Hodge–Korn completions; determinants and minor charging; finite heads and adaptive filtering; spectral and operator-theoretic detectors; realization and factorization rigidity; and the semilocal branch-selection coordinate. Each entry is classified as a no-go theorem inside the literal Riemann system, a conditional falsifier valid under the off-line counterfactual, a countermodel satisfying all structural hypotheses of a proposed mechanism, or a scale obstruction.

Three structural principles account for every entry. *Radical saturation*: the infinite-dimensional family $K^{(2j)}/K$ attains equality, and its first member is unbounded, so no decomposition into nonnegative summands may contain a local edge square. *Measure capacity*: the source carries singular mass on the lines $t \pm s = \log p^k$ while the polar pair law is absolutely continuous, so no endpoint-preserving positive coupling exists. *Counterfactual invariance*: under the existence of an off-line orbit, every positive datum of the system persists while the strict surplus fails, so no implication insensitive to the location of the mean-periodic frequencies can succeed.

We also record the unconditional structure that survives, including an exact and fully explicit Gamma absorption theorem $2K(u) < \frac{1501}{2000} r_\Gamma(u)$ which removes the Hermitian connection power with a fixed margin, and a rigidity theorem showing that the remaining contraction threshold is invariant under every change of signed realization. We conclude by delimiting the narrow corridor of mechanisms that remain admissible. No statement in this paper proves the Riemann Hypothesis, and by the counterfactual invariance principle none of them could.

CONTENTS

1.	Purpose	2
1.1.	What is not claimed	2
2.	The canonical coordinate	2
2.1.	The radical	3
3.	Unconditional structure that survives	4
3.1.	The Gamma absorption theorem	4
3.2.	The common PNT Stieltjes form	5

Independent Researcher. dtrejopizzo@gmail.com.

4. The circularity boundary	6
5. Taxonomy	7
6. Family A. Order, positivity preservation, total positivity	7
7. Family B. Coupling, transport, martingale routing	8
8. Family C. Local squares, sums of squares, Hodge–Korn	8
9. Family D. Determinants, minors, charging	10
10. Family E. Finite heads, staircases, adaptive filtering	11
11. Family F. Spectral and operator-theoretic detectors	12
12. Family G. Realization and factorization rigidity	14
13. Family H. The semilocal branch-selection coordinate	15
14. Three structural principles	16
15. The admissible corridor	16
16. Summary table	17
17. Conclusion	18
References	19

1. PURPOSE

A research programme that reduces a famous conjecture to a single inequality produces two kinds of output. The first is the reduction itself. The second, usually discarded, is the record of everything that was tried against the surviving inequality and provably cannot work.

The second output is the subject of this paper. Our reason for publishing it separately is stated precisely in Section 4: in the coordinate developed here, the surviving inequality is not a preliminary estimate from which Weil positivity follows. It *is* Weil positivity, evaluated at a stationary residual. Consequently a catalogue of obstructions is not a consolation prize for having failed to close the gate. It is the part of the work whose validity does not depend on the gate ever being closed, and it is directly useful to anyone else approaching the same wall.

We have tried to make each entry falsifiable in the ordinary sense: it either exhibits an explicit object in the literal Riemann system, or it exhibits a model satisfying a stated list of hypotheses and violating the proposed conclusion. Where an entry rules out a mechanism we ourselves proposed in earlier work, we say so.

1.1. What is not claimed. We do not claim that the Riemann Hypothesis is false, nor that the Connes–CCM route is unviable. Every entry below rules out a *method*, not the conclusion. We also do not claim the catalogue is exhaustive; it is exhaustive only over the mechanisms we tested.

2. THE CANONICAL COORDINATE

All obstructions are stated in one coordinate, so that they can be compared. We recall its construction.

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(z) = \xi(\frac{1}{2} + iz)$. Let K be Riemann's even kernel, normalized by

$$(2.1) \quad \widehat{K} = \Xi.$$

The theta representation gives, for $x \geq 0$,

$$(2.2) \quad K(x) = 2\pi e^{5x/2} \sum_{m \geq 1} m^2 (2\pi m^2 e^{2x} - 3) e^{-\pi m^2 e^{2x}},$$

in which every summand is positive because $2\pi m^2 e^{2x} > 3$. Hence K is smooth, even, strictly positive and double-exponentially decreasing. Put

$$(2.3) \quad h(x) = \cosh(x/2), \quad c_K = \int_{\mathbb{R}} h(x) K(x) dx = \widehat{K}(i/2) = \xi(0) = \frac{1}{2}.$$

Theorem 2.1 (Full-kernel Doob–variance identity). *For every even compactly supported smooth multiplier r ,*

$$(2.4) \quad \boxed{\mathcal{Q}_W(Kr, Kr) = \mathcal{E}_K(r) - \frac{1}{2} \text{Var}_{\mu_K}(r),}$$

where

$$(2.5) \quad \mathcal{E}_K(r) = \int_0^\infty \int_{\mathbb{R}} K(x) K(x-u) |r(x) - r(x-u)|^2 dx d\nu_\zeta(u),$$

$$(2.6) \quad d\nu_\zeta(u) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \delta_{\log n}(du) + \frac{e^{-u/2}}{1 - e^{-2u}} du, \quad d\mu_K = \frac{hK dx}{c_K}.$$

The identity is unconditional. Consequently the Riemann Hypothesis is equivalent to the single sharp Poincaré inequality

$$(2.7) \quad \boxed{\mathcal{E}_K(r) \geq \frac{1}{2} \text{Var}_{\mu_K}(r)}$$

for all even compactly supported smooth r .

The virtue of (2.4) is that it is an identity, not an estimate: no prime, Gamma or polar term has been separated, bounded, or discarded. Every obstruction below is therefore an obstruction to proving (2.7), and can be compared with every other. We write $\mathcal{J}_u(r)$ for the inner displacement form in (2.5),

$$(2.8) \quad \mathcal{J}_u(r) = \int_{\mathbb{R}} K(x) K(x-u) |r(x) - r(x-u)|^2 dx \geq 0,$$

so that $\mathcal{E}_K(r) = \int_0^\infty \mathcal{J}_u(r) d\nu_\zeta(u)$.

2.1. The radical. Let $L \geq 0$ be the self-adjoint generator associated with the closed form $\mathcal{E}_K$ on $L^2_{\text{even}}(\mu_K)$, and set

$$(2.9) \quad \mathcal{R} = \overline{\text{span}} \left\{ \frac{K^{(2j)}}{K} : j \geq 1 \right\}_{\text{centered}}, \quad \mathcal{C} = (\mathbf{1} \oplus \mathcal{R})^\perp.$$

Then $L\mathbf{1} = 0$ and $Lr = \frac{1}{2}r$ for $r \in \mathcal{R}$: the radical is exactly the threshold eigenspace. Equivalently, in multiplier coordinates, $\mathcal{R}^\perp$ is characterized by the convolution equation

$$(2.10) \quad (hq) * K = 0,$$

whose analytic solutions are mean-periodic with frequencies in the zero divisor of Ξ , including the explicit modes $\chi_z(x) = \cos(zx)/h(x)$. We write $\mathcal{N}_K = \{F : F * K = 0\}$.

3. UNCONDITIONAL STRUCTURE THAT SURVIVES

Before the catalogue we record what is proved, since several obstructions are sharp precisely because these positive facts are available.

Theorem 3.1 (Essential threshold). $\sigma_{\text{ess}}(L|_{1^\perp}) \subset [1/2, \infty)$. Hence, after shorting the exact threshold radical, every spectral point of $L|_{\mathcal{C}}$ in $(0, 1/2)$ is an isolated eigenvalue of finite multiplicity.

The proof combines local compactness of the form domain, obtained from the Gamma small-jump estimate $\mathcal{E}_\Gamma + C\|f\|^2 \gtrsim \int \log(2 + |\xi|)|\hat{f}(\xi)|^2 d\xi$, with a moving prime-number-theorem quadrature giving the exact tail mass $2c_K^2 = 1/2$.

Theorem 3.2 (Heat-core Galerkin exhaustion). *The form $\mathcal{E}_K$ is closed on its maximal domain. With $S_{\mathcal{C}} = L|_{\mathcal{C}} + \frac{1}{2}I \geq \frac{1}{2}I$ and any Hilbert-dense $(g_j) \subset \mathcal{C}$, the nested spaces $V_M^{\text{heat}} = \text{span}\{e^{-S_{\mathcal{C}}/k}g_j : 1 \leq j, k \leq M\}$ form a form core. On every fixed finite-dimensional $E \subset D(\mathcal{E}_K)$ the omitted Euler tail satisfies $\|N_E^{-1/2}(\mathcal{T}_X)_E N_E^{-1/2}\| \rightarrow 0$. A finite omitted block containing two distinct primes has positive definite Gram form on every finite centered space.*

The last assertion uses only that $\log p / \log r \notin \mathbb{Q}$, so the group generated by the two displacements is dense; a vector killed by both is invariant under a dense subgroup, hence constant, hence zero after centering. Theorem 3.2 removes weighted spectral synthesis of the zero modes as a separate hypothesis.

Theorem 3.3 (Bounded prime generator). *The literal ordinary-prime jump generator is bounded on the Doob space. Consequently every normalized subthreshold eigenstate of the complete prime-Gamma operator lies in the Gamma operator domain and has a uniform local second logarithmic Fourier moment.*

3.1. The Gamma absorption theorem. The following is the sharpest unconditional gain currently available and is used in Entries C5 and G2 below. Set

$$(3.1) \quad r_\Gamma(u) = \frac{e^{-5u/2}}{1 - e^{-2u}} \quad (u > 0), \quad a = \frac{K}{h}, \quad \mathcal{C} = T_{K'+K/2}M_a,$$

where $\mathcal{C}$ is the physical Abel connection of the delay-bank realization. For a nonnegative even displacement density v write

$$(3.2) \quad \mathfrak{b}_v(q) = \frac{1}{2} \iint_{\mathbb{R}^2} v(x-y)K(x)K(y)|q(x)-q(y)|^2 dx dy.$$

Theorem 3.4 (Pointwise theta-Gamma margin). *For all $u > 0$,*

$$(3.3) \quad \boxed{2K(u) < \frac{1501}{2000} r_\Gamma(u)}, \quad \text{hence} \quad w_\Gamma := r_\Gamma - 2K \geq \frac{499}{2000} r_\Gamma > 0.$$

Proof. Put $t = e^{2u} \geq 1$. From (2.2) and (3.1),

$$(3.4) \quad \frac{2K(u)}{r_\Gamma(u)} = 4\pi t^{3/2}(t-1) \sum_{m \geq 1} m^2 (2\pi m^2 t - 3) e^{-\pi m^2 t}.$$

Write $R_1(t) = 4\pi t^{3/2}(t-1)(2\pi t-3)e^{-\pi t}$ for the first atom. Its logarithmic derivative $d(t) = \frac{3}{2t} + \frac{1}{t-1} + \frac{2\pi}{2\pi t-3} - \pi$ satisfies $d'(t) < 0$, so R_1 has a single global maximum. Using $\frac{333}{106} < \pi < \frac{355}{113}$ and monotonicity of d in π , rational arithmetic gives $d(169/100) > 0 > d(17/10)$, so the maximizer lies

in $(169/100, 17/10)$. On that interval $t^{3/2} < \frac{17}{10} \cdot \frac{131}{100}$ since $(131/100)^2 > 17/10$; also $t - 1 < 7/10$ and $e^{-\pi t} < e^{-56277/10600} < 1/202$. Hence

$$(3.5) \quad R_1(t) < 4 \cdot \frac{355}{113} \cdot \frac{17}{10} \cdot \frac{131}{100} \cdot \frac{7}{10} \left(2 \cdot \frac{355}{113} \cdot \frac{17}{10} - 3 \right) \frac{1}{202} = \frac{240179723}{322417250} < \frac{3}{4}.$$

For $m \geq 2$, discard -3 and use $\frac{(m+1)^4 e^{-\pi(m+1)^2 t}}{m^4 e^{-\pi m^2 t}} \leq (\frac{3}{2})^4 e^{-5\pi} < 10^{-5}$ to get $\sum_{m \geq 2} m^4 e^{-\pi m^2 t} < 17e^{-4\pi t}$. With $s = t - 1$ and $(1+s)^{5/2} \leq e^{5s/2}$,

$$(3.6) \quad t^{5/2}(t-1)e^{-4\pi t} \leq \frac{e^{-4\pi}}{e(4\pi - 5/2)}.$$

Using only $\pi^2 < 10$, $\pi > 3$, $e > 2$ and $e^{4\pi} > e^{12} > 160000$, the tail contributes less than $1360/(19 \cdot 160000) < 1/2000$. Adding (3.5) proves (3.3). $\square$

Remark 3.5. The proof uses no floating-point input. The true maximum of (3.4) is ≈ 0.71779 , attained at $t \approx 1.69610$, so the stated constant has genuine slack; the tightest step is $e^{-\pi \cdot 169/100} < 1/202$, where the true value is $1/202.24$. Any application needing only a constant below 1 may replace $3/4$ by $4/5$ throughout and avoid that step.

Theorem 3.6 (Exact Gamma-connection factorization). *Let q lie in the common form core and put $F = hq$, $f = Kq$. Then*

$$(3.7) \quad \mathfrak{b}_{\Gamma,*}(q) + 2 \operatorname{Re} \langle F, \mathcal{C}F \rangle_{\omega_K} = \mathfrak{b}_{w_\Gamma}(q) + 2 \int_{\mathbb{R}} (K * K)(x) K(x) |q(x)|^2 dx.$$

Both terms on the right are nonnegative, and consequently

$$(3.8) \quad \mathfrak{b}_{\Gamma,*}(q) + 2 \operatorname{Re} \langle F, \mathcal{C}F \rangle_{\omega_K} \geq \frac{499}{2000} \mathfrak{b}_{\Gamma,*}(q).$$

Proof. Since $r_\Gamma = w_\Gamma + 2K$ we have $\mathfrak{b}_{\Gamma,*} = \mathfrak{b}_{w_\Gamma} + 2\mathfrak{b}_K$. The adjoint computation for $\mathcal{C}$ together with $c_K = 1/2$ gives $\operatorname{Re} \langle F, \mathcal{C}F \rangle_{\omega_K} = \langle f, T_K f \rangle_2$. Expanding the difference square in $\mathfrak{b}_K$ and using $\int K(x-y)K(y) dy = (K * K)(x)$ yields the exact identity

$$(3.9) \quad \mathfrak{b}_K(q) + \langle f, T_K f \rangle_2 = \int_{\mathbb{R}} (K * K)(x) K(x) |q(x)|^2 dx.$$

Combining the three displays proves (3.7); Theorem 3.4 and monotonicity of $v \mapsto \mathfrak{b}_v$ give (3.8). $\square$

The argument is performed entirely at the difference-form level and never splits the singular Gamma kernel into separately divergent diagonal and convolution pieces. Its consequence is that the Hermitian part of the connection is no longer an open sign: it is absorbed by the completed Gamma remainder with the explicit margin $499/2000$, on every heat row, every hybrid row, and indeed the full common core. The K' part of $\mathcal{C}$ is exactly skew-adjoint and therefore lossless.

3.2. The common PNT Stieltjes form. Let

$$(3.10) \quad \mathcal{D}(u) = \sum_{\log n \leq u} \frac{\Lambda(n)}{\sqrt{n}} - 2(e^{u/2} - 1), \quad k_D(u) = e^{-u/2} \{\psi(e^u) - e^u + 1\}.$$

Proposition 3.7 (Common primitive). *As distributions on $(0, \infty)$, $d\mathcal{D} = dk_D + \frac{1}{2}k_D(u) du$. Consequently the outgoing prime-number-theorem quadrature and the incoming Abel flux are the two halves of a single Stieltjes form,*

$$(3.11) \quad \boxed{\mathfrak{P}_{\text{PNT}}(q) = \int_{[0, \infty)} \mathcal{J}_u(q) d\mathcal{D}(u) = - \int_0^\infty k_D(u) \{ \mathcal{J}'_u(q) - \tfrac{1}{2} \mathcal{J}_u(q) \} du.}$$

Together with Theorem 3.6, the exact remaining physical form is

$$(3.12) \quad \mathfrak{P}_{\text{PNT}}(q) + \mathfrak{b}_{w_\Gamma}(q) + 2 \int_{\mathbb{R}} (K * K) K |q|^2 dx \geq 0,$$

with the simpler sufficient version

$$(3.13) \quad \boxed{\mathfrak{P}_{\text{PNT}}(q) + \frac{499}{2000} \mathfrak{b}_{\Gamma,*}(q) \geq 0.}$$

Entry C5 shows that (3.12)–(3.13) cannot be proved by any displacement-local mechanism.

4. THE CIRCULARITY BOUNDARY

This section states the organizing theorem of the paper. It explains why the catalogue is the correct output, and it is also the sharpest single constraint on any future attempt.

Theorem 4.1 (Off-line anti-surplus). *Assume Ξ has an off-line zero orbit. Then there is a real even $F \in \mathcal{N}_K$ in the common form domain, $q = F/h$, such that for every finite prime-power head X :*

- (1) *the finite-head pivot is negative, $\mathcal{A}_X(q, q) = -\delta_X < 0$;*
- (2) *the exact adaptive gain is the complete omitted-prime response, $G_X = \mathcal{T}_X(q, q)$;*
- (3) *the source surplus is strictly negative,*

$$(4.1) \quad \boxed{G_X - \delta_X = \mathcal{A}_\infty(q, q) < 0;}$$

- (4) *for every finite tail cutoff $Y > X$ the physical bordered determinant is negative, $\mathfrak{C}_X(Y) - \delta_X < 0$.*

*All displacement forms here are the literal maps $q \mapsto \sqrt{K(x)K(x - \log n)} \{q(x) - q(x - \log n)\}$ with the ordinary weights $\Lambda(n)/\sqrt{n}$, with the complete Gamma channel, the polar threshold, and the equation $(hq) * K = 0$ all retained jointly.*

Proof. An off-line orbit produces a real even $F \in \mathcal{N}_K$ with $\mathcal{A}_\infty(q, q) < 0$. The exact decomposition $\mathcal{A}_\infty = \mathcal{A}_X + \mathcal{T}_X$ and nonnegativity of every omitted atom give (1). In the scalar row there is no old-mode regression and no radical shorting, so the resolvent defining the adaptive gain is the identity, giving (2); adding the two identities gives (3). Nonnegativity of the omitted atoms gives $\mathfrak{C}_X(Y) \leq G_X$, and with no nuisance coordinate the deleted principal determinant is 1, which gives (4). $\square$

Corollary 4.2 (Counterfactual invariance). *The positive data*

$$\Lambda(p^k) > 0, \quad K > 0, \quad \mathcal{J}_{\log p^k}(q, q) \geq 0, \quad (hq) * K = 0$$

all hold for the vector constructed under the off-line counterfactual, while the strict surplus $G_X > \delta_X$ does not. Hence no implication which is insensitive to the location of the mean-periodic frequencies can prove the surplus.

Remark 4.3 (The surplus is not a lemma). Identity (4.1) says that the desired surplus is not a preliminary estimate from which completed positivity follows. It is completed positivity, evaluated at the stationary residual. Any proposed proof should therefore be audited at the first line where (4.1) is assigned a positive sign; that line carries the entire Riemann-strength input. This is the single most useful practical consequence of the present work.

5. TAXONOMY

Each entry below carries one of four tags.

- [**NO-GO**]: A theorem in the literal Riemann system showing that a class of arguments cannot produce (2.7). Strongest form.
- [**FALSIFIER**]: A construction, valid under the off-line counterfactual and inside the literal system, on which the proposed inequality is false. By Corollary 4.2 such an entry shows the mechanism cannot be counterfactual-insensitive.
- [**COUNTERMODEL**]: An explicit model satisfying every structural hypothesis of a proposed mechanism and violating its conclusion. Rules out proofs using only the listed hypotheses; does not itself concern ζ .
- [**SCALE**]: A quantitative envelope mismatch: the mechanism is valid but provably loses more than the quantity it must control.

We emphasize the difference between [NO-GO] and [COUNTERMODEL]. A countermodel rules out an *axiomatization*: it shows that the properties invoked are insufficient. It leaves open that the actual Riemann system has additional properties which close the argument. A no-go rules out the mechanism in the actual system.

6. FAMILY A. ORDER, POSITIVITY PRESERVATION, TOTAL POSITIVITY

Entry 6.1 ([NO-GO] Failure of the first Beurling–Deny criterion). In the completed source formula, let $\varphi_1, \varphi_2 \geq 0$ be disjointly supported bumps whose separation d avoids every prime-power logarithm. Their cross form is strictly positive: the polar kernel grows like $e^{d/2}$ while the opposing Gamma kernel decays like $e^{-d/2}$. Hence the completed heat semigroup is not positivity preserving, in the full sector and in the even sector.

Consequence. No Perron–Frobenius, Harnack, or ground-state domination argument based on the natural pointwise cone applies to the completed operator. This is significant because the ground-state heuristic — “the bottom eigenvector should be positive, hence the radical should be the bottom” — is the most natural first attack, and it is unavailable.

Entry 6.2 ([NO-GO] The system is not TP_2). The exact Gamma density g is strictly log-convex, so $g(t)^2 - g(t-\varepsilon)g(t+\varepsilon) < 0$. Hence neither the Gamma kernel nor the short-time full semigroup is totally positive of order two.

Consequence. Variation-diminishing spectral ordering is unavailable. In particular one cannot conclude from the fact that the threshold state K''/K has a single sign change that it is the first centered mode.

Entry 6.3 ([NO-GO] Failure of stochastic monotonicity). The folded ordinary-prime–Gamma Doob process has, for each displacement u , the reflected branch $t \mapsto |t-u|$. With the literal atom $p = 2$, at $t_0 = \frac{1}{4} \log 2$ and $z = \frac{3}{4} \log 2$, the upper-tail jump rate has a downward discontinuity of exactly $c_K \Lambda(2)K(z)/(\sqrt{2}h(t_0)) > 0$; the Gamma continuum and every other prime-power atom

form a continuous remainder because no other displacement meets that boundary. This violates the necessary tail-rate order.

Consequence. Both the short-time semigroup and the large-parameter resolvent reverse the increasing cone on explicit smooth generator tests. Ordering K''/K as the first centered mode by stochastic monotonicity or by an invariant increasing cone is impossible. Together with A2, the entire “order” family is closed.

7. FAMILY B. COUPLING, TRANSPORT, MARTINGALE ROUTING

Entry 7.1 ([NO-GO] No radical-sharp first-chaos coupling). Restore the continuous theta coordinate and the integer and rational theta marks in all four channels (Gamma, divisible, nondivisible fractional, central crossing), giving an exact positive latent Dirichlet form with the literal coefficients. No positive conditional-expectation or first-chaos projection maps this lift to the polar pair gradient while preserving the full equality family $\{K^{(2j)}/K\}$.

Mechanism. Equality in conditional Jensen makes every radical edge signature deterministic; the signatures separate oriented endpoints and therefore force endpoint preservation. But the literal prime edges carry singular source mass on the lines $t - s = \log n$ and $t + s = \log n$, while the polar pair law is absolutely continuous. This is an exact measure-capacity contradiction. A second, independent capacity obstruction holds in the Gamma channel alone: its density lacks polar capacity on remote fixed-displacement rectangles.

Entry 7.2 ([NO-GO] No positive martingale routing). In the folded polar theta law with dilation variable $S = Be^T$, choosing a prime-power divisor $N \mid b$ with probability $\Lambda(N)/\log b$ identifies the divisor current and the cross-divisor dispersion as the conditional mean and variance of one increment, and the resulting ANOVA transform is unitary. Nevertheless every finite or countable positive martingale routing retaining all currents, dispersions, fractional and central fibres, and Gamma, obeys an exact nonnegative defect identity; radical equality forces every martingale difference and every unused source square to vanish, hence endpoint preservation, hence the contradiction of B1. The natural separated assignment already fails on the two-state fibre $2 < S < 3$.

Entry 7.3 ([NO-GO] Shorting is not a weakening). Polarization shows that shorting subtracts exactly the same radical square from the source and from the target. Hence the complementary inequality is *equivalent* to, not weaker than, the original all-prime inequality. Moreover radical completeness is refuted exactly: for every zero z , the mode $\chi_z(x) = \cos(zx)/\cosh(x/2)$ is a nonzero $L^2(\mu_K)$ element orthogonal to all $K^{(2j)}/K$.

Consequence. One cannot buy room by passing to the radical complement. Every reformulation obtained by shorting inherits the exact original difficulty. This entry is used repeatedly below.

8. FAMILY C. LOCAL SQUARES, SUMS OF SQUARES, HODGE–KORN

This family contains the master obstruction of the paper.

Entry 8.1 ([NO-GO] Radical saturation: no local positive remainder). The family $r_j = K^{(2j)}/K$, $j \geq 1$, satisfies

$$(7.1) \quad \mathcal{E}_K(r_j) = \frac{1}{2} \text{Var}_{\mu_K}(r_j) \quad (j \geq 1),$$

so the sharp inequality (2.7) is saturated on an infinite-dimensional space. Suppose a proposed proof has the form

$$(7.2) \quad \mathcal{E}_K(r) - \frac{1}{2} \text{Var}_{\mu_K}(r) = \sum_{\alpha} \mathcal{S}_{\alpha}(r), \quad \mathcal{S}_{\alpha} \geq 0,$$

and that one summand dominates a local edge square,

$$(7.3) \quad \mathcal{S}_{\alpha_0}(r) \geq \int_0^{\infty} \int_{\mathbb{R}} w(x, u) |r(x) - r(x - u)|^2 dx du$$

with $w \geq 0$ positive on a set of positive measure. Then (7.2) is impossible.

Proof. Take $r_1 = K''/K$. By (7.1) the left side of (7.2) vanishes, so every nonnegative summand vanishes; (7.3) then forces $r_1(x) = r_1(x - u)$ for almost every (x, u) in the positive set of w . As r_1 is real analytic, Fubini and analytic continuation make it periodic for at least one nonzero displacement. But the leading positive theta term gives $K''(x)/K(x) \asymp e^{4x}$ as $x \rightarrow +\infty$, so r_1 is unbounded and cannot be periodic. $\square$

Consequence. The following are all excluded:

- (1) canonical-path comparison using only divisible theta indices;
- (2) Efron–Stein decomposition over individual prime towers;
- (3) any Cheeger estimate with a strict edge surplus;
- (4) any sum-of-squares proof whose individual squares are local translation differences.

What survives is a *signed* global decomposition with genuine cancellation between different prime channels. Entry C1 is the reason every later entry in this paper insists on the word “signed”.

Entry 8.2 ([NO-GO] Finite-incidence Hodge–Korn rigidity). The vertex signatures $(K^{(2j)}(x)/K(x))_{j \geq 0}$ are finitely linearly independent. Hence a local linear combination of edge increments which vanishes on the complete radical is divergence free; it is an incidence cycle and therefore vanishes on every gradient. Since the theta-divisor fibres are stars, every radical-sharp local amplitude on their current, dispersion or residue-character coordinates is zero.

Consequence. No nontrivial integral of finite-local squares can complete the signed Gamma–polar channel. After the exact global radical anti-short one obtains the sharp identity $\mathcal{Q}_W = \|G_{\perp} r\|^2 - \|D_{\perp} r\|^2$, whose remaining Korn contraction is globally nonlocal and is violated by a hypothetical subthreshold mode by the factor $(2\alpha)^{-1/2} > 1$.

Entry 8.3 ([NO-GO] No infinite boundary escape). In the literal source norm, with every von Mangoldt and theta weight retained, the conductance of the folded spatial cut at R is $O(\exp\{-ce^R\})$. Compact smooth even multipliers form a core of the closed prime–Gamma form, so every bounded locally divergence-free current is orthogonal to all form-domain gradients and has zero flux at the folded end. A nonzero cofinal boundary flux would require squared source norm at least $\exp\{ce^R\}$.

Consequence. C2 cannot be evaded by moving the missing amplitude to infinity. There is no extra bounded boundary-homology sector.

Entry 8.4 ([NO-GO] Primitive and Volterra squares are indefinite). Pushing the identities $k_y = \frac{1}{2}(D^2 - \frac{1}{4})f_y$ and $\phi(y) = -\pi(y^3 e^{-\pi y^2})'$ through the complete source, the multiplicative pushforward is the single literal signed measure

$$(7.4) \quad d\eta = \sum_{n \geq 2} \Lambda(n) \delta_n + \left\{ -1 + \frac{1}{z(z^2 - 1)} \right\} dz.$$

Fibrewise positivity is false: before the first prime, η is strictly negative on $(4/3, 2)$, and even the quadratic increment satisfies $A_{(\log z)^2}(2^-) < -\frac{43}{4500}$. Summing every theta and character fibre gives exactly $F_r(z) = z^{-1/2} J_{\log z}(r)$, whose cumulative rises to the plastic root, then has strictly negative continuous drift between prime powers with upward jumps $\Lambda(n)n^{-1/2} J_{\log n}$.

Consequence. Neither the fibrewise nor the fully aggregated coordinate is a local positive square. The endpoint terms vanish, so Gamma and the pole cannot be appended afterwards as a boundary correction: they are already inside the signed cumulative.

Entry 8.5 ([NO-GO] No centered-path or local Stieltjes completion). Let $C_u q = \{K(x)K(x-u)\}^{1/2}\{q(x) - q(x-u)\}$ and $Z_q(u) = e^{-u/4} C_u q$, so $\|Z_q(u)\|^2 = e^{-u/2} \mathcal{J}_u(q)$. For $0 < \theta \leq 1$ set $a_\theta(u) = \theta e^{u/2} r_\Gamma(u) > 0$. Then

$$(7.5) \quad \mathfrak{P}_{\text{PNT}} + \theta \mathfrak{b}_{\Gamma,*} = \int_0^\infty a_\theta \left\| Z_q - \frac{E_1(e^u)}{a_\theta} Z'_q \right\|^2 du - \int_0^\infty \frac{E_1(e^u)^2}{a_\theta(u)} \|Z'_q(u)\|^2 du,$$

with $E_1(T) = \psi(T) - T + 1$. The second line has the wrong sign and is not an omitted Gamma term: the entire Gamma budget is already inside a_θ . Moreover, at the available margin $\theta_* = 499/2000$, the scalar density of the sufficient form (3.13) is $w_*(u) = -e^{u/2} + \theta_* r_\Gamma(u)$, and with $z = e^u$

$$(7.6) \quad \frac{\theta_* r_\Gamma(u)}{e^{u/2}} = \frac{\theta_*}{z^3 - z}, \quad z \geq \frac{9}{8} \implies z^3 - z \geq \frac{153}{512} > \frac{499}{2000},$$

since $306000 > 255488$. Hence

$$(7.7) \quad \boxed{w_*(u) < 0 \quad (\log \frac{9}{8} \leq u < \log 2).}$$

In the exact residual (3.12) the density $-e^{u/2} + r_\Gamma - 2K$ is likewise strictly negative on $[\log \frac{3}{2}, \log 2)$, because $z^3 - z \geq \frac{15}{8} > 1$ and $K > 0$.

Quantitative version. Let $q_N = \chi \cos(Nx)$ with $0 \neq \chi \in C_c^\infty$ chosen so that its support overlaps its translates on a fixed $I \Subset (0, \log 2)$. Then $\int_I \|Z'_{q_N}(u)\|^2 du \geq c_I N^2$ for large N , whereas

$$(7.8) \quad \mathfrak{b}_{\Gamma,*}(q_N) = O(\log N), \quad \mathfrak{P}_{\text{PNT}}(q_N) = O(1).$$

The two N^2 terms in (7.5) cancel each other exactly; estimating either separately destroys the cancellation by a factor larger than any fixed Gamma margin. Projection off any fixed finite radical block changes these quantities by $O(N^{-A})$ for every A .

Consequence. None of the following can be the next step: positivity of the common displacement density; a pointwise completion of the centered Abel derivative; a bound of $\mathcal{J}'_u$ or Z'_q by the Gamma form; or a finite-radical version of any of these. Note that (7.7) locates the failure *below the first prime*, where the arithmetic is entirely explicit — the obstruction is not an artifact of prime irregularity.

9. FAMILY D. DETERMINANTS, MINORS, CHARGING

Entry 9.1 ([NO-GO] Cauchy–Binet charging is circular). In exterior-power coordinates, an abstract contractive charge exists exactly when $\tau_{d+1} \geq \delta_J \tau_d$, i.e. exactly when the inequality to be proved holds. Consequently the squared-minor expansion supplies no sign. Moreover a universal positive endpoint-local weight-preserving charge would saturate Jensen on every radical signature, hence preserve endpoints, and then contradict B1.

Consequence. The only surviving possibility is a row-dependent, globally nonlocal charge after the complete radical anti-short, whose strict norm bound is precisely the projected Plücker-frame inequality. That is the surplus again, not a route to it.

Entry 9.2 ([COUNTERMODEL] Network positivity does not force crossing). There is an exact four-state rational model with a connected full-rank tail, nontrivial adaptation, an exact threshold radical, positive Gamma and head graph channels, literal weights $\Lambda(n)/\sqrt{n}$, and a theta-sized envelope, for which the bordered determinant remains negative.

Consequence. Effective resistance, matrix-tree expansions, generic network positivity and summability do not force the crossing.

Entry 9.3 ([COUNTERMODEL] Exact saturation with the literal theta data). A one-dimensional model using the actual theta displacement forms, the actual lengths $\log p^k$, and the ordinary weights $\Lambda(p^k) = \log p$, with the free source diagonal set equal to the complete positive tail, has every proper determinant strictly below the deficit, converging to it. Permanent non-crossing is exactly $\sigma_\infty \leq 0$.

Consequence. This rules out any proof from source algebra, positive ordinary-prime increments, inertia and summability alone. Total positivity supplies no missing input either: the Gamma kernel is not TP_2 (A2), the literal prime tower has a PF_2 failure, and the polarized jet j_2 is already indefinite on $\{p, p^2\}$.

Entry 9.4 ([SCALE] The observation-only determinant is strictly lossy). The pure Gram ratio $\det G_B^{(M)} / \det G_B^{(M-1)}$ is a valid but strictly weaker certificate than the exact adaptive gain. In stable floating-point diagnostics at dimensions 4, 7, 12 it misses the initial deficit while the exact Kalman gain crosses it, sometimes by five orders of magnitude. The missing energy is the component parallel to the old observation range, priced by the positive preceding prime-Gamma block.

Consequence. The force-bearing object is the augmented determinant

$$(8.1) \quad \Delta_B = \frac{\det \begin{pmatrix} A + U^*U & U^*r \\ r^*U & \|r\|^2 \end{pmatrix}}{\det(A + U^*U)},$$

not the observation Gram ratio. We record this because the weaker certificate is the one that suggests itself first.

10. FAMILY E. FINITE HEADS, STAIRCASES, ADAPTIVE FILTERING

Entry 10.1 ([NO-GO] No finite-head telescope). Every proper finite prime-power head is strictly negative on each nonconstant exact Riemann-radical direction. Hence its signed short over the radical is unbounded below, and the full infinite prime-Gamma-polar limit must be assembled *before* radical shorting.

Consequence. A finite-head Cholesky or Pick telescope is impossible. This is the finite-level shadow of B3.

Entry 10.2 ([COUNTERMODEL] Discrete levels and full-rank sensors are not enough). There is an exact three-state reversible graph generator with a constant ground state, an exact completed $1/2$ -radical, discrete levels, and an infinite norm-convergent bank of strictly positive full-rank sensors carrying the literal weights $\Lambda(p^k)/p^{k/2}$, which nevertheless retains a complementary eigenvalue $1/4$.

Entry 10.3 ([COUNTERMODEL] Adaptation loss defeats tail energy). There is an injective two-mode theta-scale model whose residual tail energy strictly exceeds its deficit, but which loses enough energy to adaptive regression that its completed pivot remains negative.

Consequence. Schur adaptivity, positive literal weights, strict observability and summability do not force crossing. The missing input is an estimate of the adaptation-loss term itself.

Entry 10.4 ([COUNTERMODEL] Exact equality saturation over the prime powers). There is a saturation model indexed by the ordinary prime powers, with the literal weights $\Lambda(m)/\sqrt{m}$, an $e^{-2\pi m}$ theta envelope, injective atom features and zero adaptive leakage, whose cumulative gain equals δ exactly: every finite pivot is negative and the completed pivot is zero.

Consequence. Stieltjes positivity and theta summability perform the finite selection only *after* a strict surplus has been proved. Equality does not suffice. This entry is the reason the target inequality must be strict, and the reason a compensated vanishing-tolerance form is needed for the limit.

Entry 10.5 ([NO-GO] Response–inertia bound). For a Hermitian H and any T , $\nu_-(H|_{\ker T}) \geq \nu_-(H) - \text{rank } T$. Hence whenever a signed finite spectral block has more negative directions than available prime-midpoint rows, some negative vector annihilates every one of those rows.

Consequence. No arbitrary-vector midpoint frame floor may be used. For the actual staircase the dimension obstruction disappears, because $A \succ 0$ with a negative Schur pivot forces negative index exactly one; but no inertia identity then bounds the regression residual from below.

Entry 10.6 ([NO-GO] Local observability does not pay the deficit). On every finite analytic even mode space the literal displacement feature restricted to any nonempty midpoint aperture is injective, giving the computable Christoffel bound $\mathcal{J}_{u,W}(q^*) \geq \kappa_M(u, W) > 0$. However, for a nonzero completed radical vector the finite-head deficit equals the sum of *all* omitted positive atom energies. Hence every proper finite omitted block gains strictly less than the deficit, and its moving-midpoint mass can tend to zero.

Consequence. Any surviving estimate must use the completed anti-short $q^* \in \mathcal{R}^\perp$ quantitatively; local observability, the adaptive source equation, and tail localization are individually insufficient.

Entry 10.7 ([SCALE] Prime-sampling envelope mismatch). On every fixed spectral block $|\text{Im } z| \leq b < 1/2$ with jets of order at most J , and every fixed $\delta > 0$,

$$(9.1) \quad \sum_{X < p \leq (1+\delta)X} \beta_p |A_p(q) + \rho_p(q)|^2 \geq c'_{\mathcal{Z},\delta} X^{3-b} (1 + \log X)^{-2J} e^{-2\pi(1+\delta)X} \|q\|_{\text{coeff}}^2.$$

This is a genuine quantitative sampling theorem, proved from the prime number theorem, but compared with the natural tail scale $e^{-2\pi X}$ it loses $e^{-2\pi\delta X}$.

Consequence. A relative-interval attack cannot place the response inside the $O(1)$ prime window where the theta weight is comparable to its leading tail value. Remez and Turán localization on the same relative interval do not repair this. Any successful estimate must be a short-window statement.

11. FAMILY F. SPECTRAL AND OPERATOR-THEORETIC DETECTORS

Entry 11.1 ([NO-GO] Birman–Schwinger supplies no sign). A compactly supported positive boost produces a compact Birman–Schwinger operator $\mathcal{K}_{\lambda,R,M}$ whose eigenvalue 1 detects each

isolated subthreshold bound state. But the exact Rayleigh quotient shows that $\|\mathcal{K}_{\lambda,R,M}\| \leq 1$ is, term for term, equivalent to $\mathcal{E}_K \geq \lambda \|\cdot\|^2$.

Consequence. Localization proves Fredholm compactness and supplies no sign. This is the faithful nonlocal version of the earlier Birman–Schwinger wall, not a new mechanism.

Entry 11.2 ([NO-GO] Mourre and dilation commutators). After ground-state conjugation, the translation commutator is parity odd and has identically zero quadratic form on the physical even sector. Dilation is parity even, but every prime power at $a = \log n$ contributes the isolated order-one distribution $a(\delta'_a - \delta'_{-a})$, and the complete raw dilation commutator is unbounded in both signs on compact oscillatory even rows. In the exact complement, translation is tangent to $F * K = 0$ but changes parity, while dilation preserves parity and has transverse leakage $t\Xi'(t)\widehat{F}(t)$ at every simple zero. Below the essential threshold a strict projected Mourre estimate holds exactly when the corresponding spectral projection is zero, and the normalized commutator vanishes on heat or hybrid rows concentrating on a bound cluster.

Entry 11.3 ([NO-GO] Heat moments and operator-monotone transforms). Every raw heat Hankel matrix is automatically positive, whereas eventual positivity of any one odd moment shifted to the threshold is already equivalent to the physical surplus. The same wall persists after every strictly increasing operator-monotone transform.

Consequence. Heat summability and scalar moment positivity supply no hidden fraction of the missing bound. Relatedly, the relative-heat observable has logarithmic decay rate exactly $-\{\inf \sigma(L|_{\mathcal{C}}) + \frac{1}{2}\}$, so the desired surplus is equivalent to rate at most -1 while the unconditional energy gives only $-1/2$.

Entry 11.4 ([NO-GO] Positive source atoms tolerate a subthreshold mode). If a subthreshold ground cluster $\alpha < 1/2$ of multiplicity m_α exists, the Birman–Solomyak relative measure has exact local density $m_\alpha d\lambda$, so $\Theta_V(t) \sim m_\alpha e^{-t(\alpha+1/2)}$ for every positive injective trace-class boost, and the heat-weighted source satisfies $\frac{1}{2}A_V - E_V \sim (\frac{1}{2} - \alpha)m_\alpha t^{-1}e^{-t(\alpha+1/2)} > 0$.

Consequence. Positive source atoms and all unshifted heat moments remain compatible with a subthreshold mode. A closure must estimate the *compensated* source itself.

Entry 11.5 ([SCALE] Non-uniformity as the deficit vanishes). For every fixed $\delta > 0$ the subthreshold spectral projection below $1/2 - \delta$ is finite rank, and one finite heat-core compression detects every such eigenstate below $1/2 - \delta/2$. The reduction is not uniform as $\delta \downarrow 0$: packets of amplitude $(\log N)^{-1/2}$ have vanishing Hilbert mass at frequency N but fixed Gamma form energy, and Hilbert accuracy $\eta^2 \asymp \delta$ already requires $\Omega \gtrsim \exp(C_R/\delta)$.

Consequence. Fixed-gap violations genuinely reduce to finite arithmetic; near-threshold violations do not. Any finite-dimensional computational strategy must therefore come with a proof that the deficit is bounded away from zero, which is itself part of the problem.

Entry 11.6 ([NO-GO] No central tightness from mean periodicity). Multiplicity jets are essential to local Ξ -universality: a compactly supported annihilator would have an entire Fourier–Laplace transform with $O(T)$ zeros, while Riemann–von Mangoldt gives $\asymp T \log T$ zeros with multiplicity. Hence exact mean-periodic zero modes approximate zero on any central window and an arbitrary profile on a disjoint window. A diagonal construction gives normalized real-even graph-domain vectors in the exact radical complement whose mass escapes equally to both tails while every localized second-log norm tends to zero.

Consequence. Mean periodicity, evenness, and the regularity of Theorem 3.3 cannot yield central tightness. The remaining input must be a global threshold-resolvent or eigenstate-observability estimate using the literal operator equation.

12. FAMILY G. REALIZATION AND FACTORIZATION RIGIDITY

This family is the most structurally informative: it shows that the missing constant is invariant under every repackaging of the problem.

Entry 12.1 ([NO-GO] Canonical signed transfer has a universal threshold). After constants and the complete radical have been shorted, the unique exact signed transfer is $C_0 = 2^{-1/2}U_DA^{-1/2}U_A^*$, with

$$(10.1) \quad \|C_0\| = (2 \inf \sigma(A))^{-1/2}.$$

Hence every exact signed kernel has the *same* contraction threshold, and a subthreshold eigenmode is amplified by precisely $(2\alpha)^{-1/2} > 1$. A virial identity further shows that a square plus a trace-zero commutator cannot conceal such a mode.

Entry 12.2 ([NO-GO] Compact all-atom factorization is unique-minimal). Let $\tilde{\mathcal{G}}$ be the complete literal von Mangoldt–Gamma displacement gradient and $\tilde{A} = \tilde{\mathcal{G}}^*\tilde{\mathcal{G}}$ after exact anti-shortening. Both $\mathcal{J} = [T_K, M_a]$ and $\mathcal{C} = T_{K'+K/2}M_a$ are Hilbert–Schmidt on the physical mean-periodic space, with $\|\mathcal{T}_b\|_{\text{HS}}^2 = c_K^2 \iint |b(x-y)|^2 d\omega_K d\omega_K$. Moreover $\tilde{A} \succeq \alpha_0 I$ for some $\alpha_0 > 0$, by Theorem 3.1 together with triviality of the kernel after removing constants. Hence $\mathcal{C} = H_C \tilde{\mathcal{G}}$ with $H_C = \mathcal{C} \tilde{A}^{-1/2} U_A^*$ compact and of minimal norm,

$$(10.2) \quad \inf_{\mathcal{C}=\tilde{H}\tilde{\mathcal{G}}} \|H\|^2 = \|\mathcal{C} \tilde{A}^{-1} \mathcal{C}^\sharp\|, \quad \|H_C\| \leq 1 \iff \mathcal{C}^\sharp \mathcal{C} \preceq \tilde{A}.$$

These gains are operator-norm limits of common-cutoff regularized factorizations containing every prime power cofinally.

Consequence. A change of realization cannot manufacture the strict gain. Entries G1 and G2 together constitute a rigidity theorem about the entire programme: the remaining constant is not a design parameter, and any future reformulation will land on the same threshold. We regard this as the single most decision-relevant fact for anyone continuing the work.

Entry 12.3 ([NO-GO] Local passivity cells cannot be multiplied). The explicit positive periodic kernel $K_L = 1 + \varepsilon \cos(4\pi x/L) + \varepsilon \cos(6\pi x/L)$ has exact mean-periodic row $F_L = \cos(2\pi x/L)$, fixed Hermitian connection power $\varepsilon^3/16$, and Gamma plus any fixed finite bank of literal delays $(\log n, \Lambda(n)/\sqrt{n})$ of size $O(L^{-2})$. The full minimal Schur cost also stays bounded away from zero, and the witness is exactly representable on a heat or hybrid Galerkin row. A moving bank can avoid the obstruction only if its weighted second delay moment is $\gtrsim L^2$, whereas an individual phase-resolving delay has $\log n \asymp L$.

Consequence. Derivative storage cannot create the missing sign, and independently passive prime cells cannot be multiplied to obtain it. This is explicitly an abstract positive-kernel falsifier, not the theta kernel; its content is that any certificate must use the cofinal prime *placement* jointly with Gamma, theta, and the polar channel, rather than Gamma or a fixed prime head.

Entry 12.4 ([NO-GO] Nonlinear amplification is unavailable). On the exact complement $(hq) * K = 0$: translation, Möbius and Jordan multipliers preserve each frequency and cannot amplify its transverse exponent. Pointwise powers do amplify it, but an off-axis mode $\cos(zx)^k$ necessarily

leaves the zero divisor once $k|\operatorname{Im} z| > 1/2$; tensor powers retain the equations only before the non-intertwining diagonal restriction. The spatial Möbius inverse cannot be passed through the zero-mode convolution: the required squarefree absolute series diverges, and the regularized symbol is $\zeta(s)/\zeta(s + \varepsilon)$, singular at exactly the hypothetical zero.

Entry 12.5 ([NO-GO] The bar homotopy has no critical contraction). The normalized multiplicative bar homotopy $Z_\varepsilon^{-1} = \sum_{k \geq 0} (-X_\varepsilon)^k$ recovers every literal von Mangoldt weight from $Z_\varepsilon^{-1} \delta Z_\varepsilon$. If ordered factorization chains remain orthogonal, the signed collapse on an r -prime squarefree fibre has exact norm $(\sum_k k! S(r, k))^{1/2} \geq \sqrt{r!}$; if they are cancelled before the norm, the result is the Möbius inverse itself. On $q_\gamma = \cos(\gamma x)/\cosh(x/2)$, a critical zero of multiplicity m forces homotopy gain $1/|\zeta(\bar{\rho} + \varepsilon)| \asymp \varepsilon^{-m}$ and connection gain $m/\varepsilon + O(1)$. The specialized bar complex has nonzero zeroth homology.

Consequence. Gamma and the pole are analytic and nonzero at the zero, so completion preserves the residue. Supplying the zero-fibre map separately returns exactly to the canonical transfer of G1.

13. FAMILY H. THE SEMILOCAL BRANCH-SELECTION COORDINATE

The remaining entries concern the finite CCM coordinate rather than the Doob coordinate, and are included because that is where the programme began.

Entry 13.1 ([NO-GO] Fixed derivative models miss off-line channels). Every fixed span of cutoff derivatives of Riemann's full kernel is asymptotically orthogonal to a hypothetical off-line evaluation channel, since every transform in that span contains the common factor Ξ . In particular, coupling K and K' does not repair selection.

Entry 13.2 ([NO-GO] Many-vector co-Poisson frames do not improve the quotient). If $a_-(Q_L)$ is the lower frame bound of the low-mode co-Poisson/prolate family on the negative spectral space, then

$$(11.1) \quad \sum_j \|P_- A_L q_{j,L}\|^2 \geq a_-(Q_L) \sum_{\epsilon_{\ell,L} < 0} \epsilon_{\ell,L}^2,$$

so a fixed negative branch forces the aggregate overlap and residual to vanish at the same rate. The scale audit is also adverse: Gamma coercivity and the absolute $O(\lambda)$ prime-pole norm control at most an $O(Le^{C\lambda})$ -dimensional negative space, while the Slepian family has only $O(\lambda^2)$ modes.

Entry 13.3 ([NO-GO] Unquantified cyclicity diverges). Although translates and derivatives of K are cyclic in ordinary L^2 , an off-line zero $\rho = \beta + i\gamma$ forces every truncated radical approximant of a detecting test to have squared error $\gg \lambda^{-2(\beta-1/2)}$. The semilocal lower-bound cost is therefore $\gg \lambda^{2(1-\beta)}$, which diverges rather than vanishes.

Entry 13.4 ([NO-GO] A previously proposed lemma is false as stated). The divisor covariance introduced in the companion paper has Fourier symbol of order $N \log N$ at quadratic scale, whereas the original von Mangoldt jump has order $\sqrt{N}(\log N)^2$; no extraction identity connects them. Under the literal Gamma-pole-threshold interpretation, a fixed compact even core orthogonal to both q_L^+ and the pole has value $-\kappa_N + O(1) = -4\lambda + o(\lambda)$, contradicting the proposed $-O(d_8)$ floor.

Consequence. We record this entry explicitly because the lemma in question was ours. Its viable successor is the complete coupled prime-Gamma-pole complement inequality, i.e. the surplus.

Entry 13.5 ([COUNTERMODEL] Qualitative positivity improvement is insufficient). There is an explicit two-dimensional positivity-improving family with a fixed negative ground state, a strictly positive near-radical model, and overlap tending to zero at exactly the residual rate.

Consequence. Endpoint normalization is angle-blind and qualitative positivity improvement cannot supply the denominator in the anti-orthogonality quotient. A viable positivity route must prove the quantitative Harnack–tightness estimate for the actual ordinary-prime operator.

14. THREE STRUCTURAL PRINCIPLES

Every entry above is an instance of one of three principles. We state them because they are more useful than the individual entries: they predict, without further computation, whether a proposed mechanism can work.

Proposition 14.1 (P1. Radical saturation). *The equality set of (2.7) contains the infinite-dimensional family $\{K^{(2j)}/K\}_{j \geq 1}$, whose first member is real analytic and unbounded. Therefore no decomposition of $\mathcal{E}_K - \frac{1}{2} \text{Var}_{\mu_K}$ into nonnegative summands may contain a summand dominating a local edge square.*

P1 governs C1, C2, C3, C4, C5, D1 and D3. *Diagnostic:* if a proposed argument produces a nonnegative leftover supported on displacements, it is already refuted.

Proposition 14.2 (P2. Measure capacity). *The source carries positive singular mass on the lines $t \pm s = \log p^k$, while the polar pair law is absolutely continuous; and the Gamma density alone lacks polar capacity on remote fixed-displacement rectangles. Therefore no endpoint-preserving positive coupling, conditional expectation, martingale routing, or weight-preserving charge from the source to the polar gradient exists.*

P2 governs B1, B2, D1 and, through Jensen equality, the endpoint clause of D2. *Diagnostic:* if a proposed argument routes source energy to target energy through a positive kernel that respects endpoints, it is already refuted.

Proposition 14.3 (P3. Counterfactual invariance). *Under the existence of an off-line orbit, all of $\Lambda(p^k) > 0$, $K > 0$, $\mathcal{J}_{\log p^k} \geq 0$ and $(hq) * K = 0$ persist, while the strict surplus fails; and the surplus itself equals the completed form at the stationary residual. Therefore no implication insensitive to the location of the mean-periodic frequencies can prove it.*

P3 governs A1–A3, E1–E7, F1–F6, G1–G5, H1–H5, and ultimately every entry. *Diagnostic:* if a proposed argument would remain valid verbatim in a system with an off-line zero, it is already refuted. This is the cheapest available audit and should be applied first.

15. THE ADMISSIBLE CORRIDOR

Negative results are only useful if they leave something. We state what the three principles jointly permit.

- (1) The decomposition must be *signed*, with genuine cancellation between distinct prime channels (P1). Individual squares are forbidden; signed differences of squares are not.
- (2) The transfer must be *nonlocal* in the displacement variable, and must not preserve endpoints (P2). It may be row dependent.

- (3) It must be applied *after* the complete mean-periodic anti-short, and must use the actual prime-power locations *before* any norm is taken (P3). A mechanism that takes norms atom by atom has already discarded the placement information which is the only thing that distinguishes ζ from the countermodels of D2, D3, E2, E3 and E4.

Concretely, the surviving target admits two equivalent exact forms. The first is the displacement form

$$(13.1) \quad \boxed{\mathfrak{P}_{\text{PNT}}(q) + \mathfrak{b}_{w_\Gamma}(q) + 2 \int_{\mathbb{R}} (K * K)K|q|^2 dx \geq 0,}$$

with the sufficient uniform-margin version (3.13). The second is the relative operator form

$$(13.2) \quad \boxed{\mathcal{C}^\sharp \mathcal{C} \preceq \tilde{A},}$$

equivalently $\|H_C\| \leq 1$ for the minimal-norm compact factor of Entry G2. By Theorem 3.6 the Hermitian connection power in (13.1) has already been absorbed with margin 499/2000; what remains is exactly the common outgoing and incoming ordinary-prime power, which by Entry C5 may not be split, and by Entry E7 may not be estimated on a relative interval.

Remark 15.1 (Honest accounting). By Theorem 4.1 both (13.1) and (13.2) are equivalent to the Riemann Hypothesis, not weaker than it. The corridor described here is therefore a description of where a proof must live, not evidence that one exists. We state this plainly because the recurrence of the same threshold constant across a dozen independent reformulations (Entries G1, G2) is easily mistaken for convergence toward a proof, when it is in fact a rigidity phenomenon: the constant is invariant precisely because the problem has not moved.

16. SUMMARY TABLE

Entry	Mechanism excluded	Type	Principle / witness
A1	Perron–Frobenius, Harnack, pointwise cone	[NO-GO]	Beurling–Deny fails; $e^{d/2}$ vs $e^{-d/2}$
A2	Variation-diminishing ordering	[NO-GO]	Gamma density log-convex, not TP_2
A3	Stochastic monotonicity, invariant cone	[NO-GO]	$p = 2$ tail-rate jump at $t_0 = \frac{1}{4} \log 2$
B1	Conditional expectation, first chaos	[NO-GO]	P2 capacity
B2	Positive martingale routing	[NO-GO]	P2 capacity; two-state fibre $2 < S < 3$
B3	Passing to the radical complement	[NO-GO]	Shorting is an equivalence
C1	Canonical paths, Efron–Stein, Cheeger, local SOS	[NO-GO]	P1; $K''/K \asymp e^{4x}$
C2	Local Hodge/Korn completion	[NO-GO]	P1; incidence cycles
C3	Boundary flux at infinity	[NO-GO]	Conductance $O(e^{-ce^R})$
C4	Gaussian/Volterra primitive squares	[NO-GO]	$\eta < 0$ on $(4/3, 2)$
C5	Centered-path, local Stieltjes	[NO-GO]	$w_* < 0$ on $[\log \frac{9}{8}, \log 2)$; N^2 loss
D1	Cauchy–Binet charging	[NO-GO]	Circular; P2 for universal charges
D2	Effective resistance, matrix-tree	[COUNTERMODEL]	ODE state rational model
D3	Source algebra, inertia, summability	[COUNTERMODEL]	ODE theta saturation model
D4	Observation-only Gram ratio	[SCALE]	Misses deficit by 10^5 at $M = 12$

Entry	Mechanism excluded	Type	Principle / witness
E1	Finite-head Cholesky/Pick telescope	[NO-GO]	Head negative on every radical direction
E2	Discrete levels + full-rank sensors	[COUNTERMODEL]	ODE state model, eigenvalue $1/4$
E3	Tail energy exceeding deficit	[COUNTERMODEL]	ODE mode adaptation-loss model
E4	Stieltjes positivity + summability	[COUNTERMODEL]	ODE equality saturation
E5	Arbitrary-vector midpoint frame floor	[NO-GO]	Response-inertia bound
E6	Local Christoffel observability	[NO-GO]	Deficit = full omitted energy
E7	Relative-interval prime sampling	[SCALE]	Loses $e^{-2\pi\delta X}$
F1	Birman–Schwinger localization	[NO-GO]	Term-for-term equivalent
F2	Mourre, dilation commutator	[NO-GO]	Odd/unbounded; vanishes on bound rows
F3	Heat moments, operator-monotone maps	[NO-GO]	Equivalent to the surplus
F4	Positive source atoms, heat moments	[NO-GO]	Compatible with $\alpha < 1/2$
F5	Finite-dimensional reduction as $\delta \downarrow 0$	[SCALE]	$\Omega \gtrsim e^{C_R/\delta}$
F6	Central tightness from mean periodicity	[NO-GO]	$T \log T$ vs $O(T)$ zeros
G1	Changing the signed kernel	[NO-GO]	Universal threshold $(2 \inf \sigma A)^{-1/2}$
G2	Changing the realization	[NO-GO]	Minimal-norm compact factor
G3	Multiplying passive local cells	[NO-GO]	K_L periodic falsifier
G4	Nonlinear amplification	[NO-GO]	Leaves divisor; $\zeta(s)/\zeta(s+\varepsilon)$
G5	Bar homotopy, Möbius inversion	[NO-GO]	ε^{-m} gain, nonzero H_0
H1	Fixed derivative models	[NO-GO]	Common factor Ξ
H2	Many-vector co-Poisson frames	[NO-GO]	Frame identity; $O(\lambda^2)$ vs $O(Le^{C\lambda})$
H3	Unquantified cyclicity	[NO-GO]	Cost $\gg \lambda^{2(1-\beta)}$
H4	Divisor-covariance extraction lemma	[NO-GO]	False as stated; $-4\lambda + o(\lambda)$
H5	Qualitative positivity improvement	[COUNTERMODEL]	ODE dimensional family

17. CONCLUSION

The companion paper reduces the Riemann Hypothesis, along the Connes–CCM–Suzuki route, to the strict physical surplus. This paper establishes three things about that surplus.

First, it is not a lemma. By Theorem 4.1 it equals the completed Weil form evaluated at the stationary residual, so a proof of it is a proof of the Riemann Hypothesis and nothing less. The practical form of this observation is Remark 4.3: audit any candidate argument at the first line where (4.1) receives a positive sign.

Second, the space of mechanisms available to prove it is much smaller than it appears. Thirty-one entries in eight families are excluded, and the exclusions are not independent: they follow from three principles — radical saturation, measure capacity, and counterfactual invariance — each of which can be checked against a proposed argument in a few lines.

Third, the missing constant is rigid. By Entries G1 and G2, every exact signed kernel and every realization of the same physical bank yields the identical contraction threshold. The repeated appearance of that constant across independent reformulations is therefore not evidence of progress toward it; it is a theorem that the problem is invariant under reformulation.

What survives is narrow and precisely stated: a signed, globally nonlocal, non-endpoint-preserving contraction applied after the complete mean-periodic anti-short, using the actual prime-power placements before any norm is taken, in either of the equivalent forms (13.1) or (13.2). The unconditional gain of Theorem 3.6 removes the Hermitian connection power from that obligation with the explicit margin 499/2000, leaving the common outgoing and incoming ordinary-prime power as the sole remaining supply.

We close with the observation that motivated this paper. A programme of this kind produces far more refutations than theorems, and the refutations are the part whose value does not depend on the conjecture ever being settled. We would have saved considerable time had a catalogue of this kind existed when we began.

REFERENCES

- [1] A. Weil, *Sur les “formules explicites” de la theorie des nombres premiers*, Comm. Sem. Math. Univ. Lund (1952), 252–265.
- [2] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [3] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022 (2026).
- [4] A. Connes, C. Consani, and H. Moscovici, *Zeta spectral triples*, arXiv:2511.22755 (2025).
- [5] A. Connes and C. Consani, *On the Jacobian of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2602.15941 (2026).
- [6] A. Connes and C. Consani, *On the Absolute Geometry of $\text{Spec } \mathbb{Z}$* , arXiv:2606.06604 (2026).
- [7] M. Suzuki, *On the Hilbert space derived from the Weil distribution*, Canadian Journal of Mathematics, FirstView (2025).
- [8] M. Suzuki, *Weil’s quadratic form via the screw function*, arXiv:2606.09096 (2026).
- [9] A. Beurling and J. Deny, *Dirichlet spaces*, Proc. Nat. Acad. Sci. U.S.A. **45** (1959), 208–215.
- [10] M. Fukushima, Y. Oshima, and M. Takeda, *Dirichlet Forms and Symmetric Markov Processes*, 2nd ed., de Gruyter, 2011.
- [11] M. Sh. Birman and M. Z. Solomyak, *Spectral Theory of Self-Adjoint Operators in Hilbert Space*, Reidel, 1987.
- [12] R. G. Douglas, *On majorization, factorization, and range inclusion of operators on Hilbert space*, Proc. Amer. Math. Soc. **17** (1966), 413–415.
- [13] E. V. Haynsworth, *Determination of the inertia of a partitioned Hermitian matrix*, Linear Algebra Appl. **1** (1968), 73–81.
- [14] S. Karlin, *Total Positivity, Vol. I*, Stanford University Press, 1968.
- [15] D. A. Trejo Pizzo, *Branch Selection in the Connes–CCM Route to RH: Weil Positivity, Prolate Zeta-Cycles, and Cofinal Prime–Gamma Inertia Matching*, companion paper (2026).